Faculty of Engineering

Module Title: Engineering Chemistry

Revision 1&2&3&4[Answered]

Chapter 1: Physical Chemistry

1. **A spray can with a volume of 0.5 L contains 2 moles of propane gas (C_3H_8) as propellant. What is the pressure, in atmospheres, of the gas in the can at 293 K?**

Answer:

PV = nRT P = nRT/ V

P = (2 ×0.0821×293) / 0.5 P = 96.26 atm

2. **Assume that you have 0.5 mol of nitrogen gas (N_2) in a volume of 0.6 L at 300 K. Calculate the pressure in atmospheres using the Van der Waals equation. [For N_2; a =1.35 L^2.atm. mol^{-2}, and b= 0.0387 $L.mol^{-1}$]**

Answer:

$$P = \frac{nRT}{V-nb} - \frac{an^2}{V^2}$$

$$P = \frac{(0.5)(0.0821)(300)}{0.6-(0.5)(0.0387)} - \frac{(1.35)(0.5)^2}{(0.6)^2}$$

P = 20.27 atm

3. **A gas has a density of 5.4 g/L at 15 °C and 1 atm. Calculate its molar mass**

Answer:

$$\rho = \frac{PM}{RT}$$

T=15+273.15=288.15 K

M = (5.4×0.082×288)/ 1

4. **A gas occupies 500 mL at 2 atm. Determine its pressure when the volume decreases to 200 mL at constant temperature.**

Answer:

$$P_1V_1 = P_2V_2$$

$$P_2 = \frac{P_1V_1}{V_2}$$

$P_2 = 2\times500/200$ $P_2 = 5$ atm

5. **A gas occupies 6.0 L at 350 K. Calculate its volume when it is cooled to 280 K at constant pressure.**

Answer:

$$V_2 = \frac{V_1T_2}{T_1}$$

$V_2 = (6\times280)/350 = 4.8$ L

6. **Complete the following statements using the appropriate words from the word bank below. Each word should be used only once.**

[Volume – Pressure – Random – Atoms – Symmetrical – Transfer - Flow – Viscosity – Below – External – Normal —Ionic]

- Boyle's law states that the volume of a gas is inversely proportional to its __**Pressure**_____ at constant temperature and amount of gas.
- Intramolecular forces are attractive forces that hold __**Atoms**__ together within a molecule.
- A nonpolar covalent bond involves the _**Symmetrical** _ distribution of shared electrons.
- An ionic bond results from the complete _**Transfer**_ of one or more electrons between atoms.
- Viscosity is a measure of a liquid's resistance to __**Flow**____.

- Liquids with stronger intermolecular forces generally have higher __**Viscosity**__ and surface tension.
- Evaporation occurs at temperatures ___**Below** _____ the boiling point of a liquid.
- The boiling point of a pure substance is reached when its vapor pressure equals the __**External**_ pressure.
- The boiling point at 1 atm is called the _**Normal**__ boiling point.

7. Write (True) or (False) for each of the following statements.

- Gases at low pressure or high temperature behave as an ideal gas **(√)**
- For an ideal gas, pressure (P) and temperature (T) is inversely proportional to each other (if all other conditions remain constant) **(√)**
- Liquids have a definite shape and an indefinite volume (×)
- Gases expand freely to fill the entire volume of their containers (×)
- According to Avogadro's law, gas volume is directly proportional to the number of moles at constant temperature and pressure (√)
- The volume occupied by gas molecules is always significant compared with the total gas volume under ideal conditions (×)
- The Van der Waals equation accounts for molecular volume and intermolecular forces (√)
- Intermolecular forces are attractive forces between molecules (√)
- A nonpolar covalent bond involves symmetrical sharing of electrons (√)
- A polar covalent bond forms between atoms with different electronegativities (√)
- An ionic bond results from the complete transfer of one or more electrons between atoms (√)
- Ion–dipole forces occur between two nonpolar molecules (×)
- Hydrogen bonding can occur between molecules containing O–H, N–H, or F–H bonds (√)
- Viscosity measures a liquid's resistance to flow (√)
- Surface tension is the resistance of a liquid to increasing its surface area (√)
- Liquids with stronger intermolecular forces generally exhibit higher viscosity and surface tension (√)
- Increasing temperature generally increases the viscosity and surface tension of liquids (×)
- Evaporation occurs only at the boiling point of a liquid (×)
- A liquid boils when its vapor pressure equals the external pressure (√)
- The normal boiling point is the boiling temperature at 1 atm (√)
- Crystalline solids have an ordered arrangement of particles extending over a long range (√)
- The melting point is the temperature at which a solid changes into a liquid (√)

8. Choose the correct answer:

- **In the ideal gas equation, R represents _____.**
  - Randomness
  - Universal gas constant
  - Rigidity
  - The amount of gas

- **STP is defined as _____ atm and _____ °C**
  - 1.0, 0
  - 1.0, 1.0
  - 273, 1.0
  - 1.0, 100

- **Which of the following physical states has a definite shape and a definite volume?**
  - Gas
  - Liquid
  - Solid
  - Vapor

- **Under which conditions does a gas tend to behave most ideally?**
  - Low temperature and high pressure
  - High temperature and low pressure
  - Low temperature and low pressure
  - High temperature and high pressure

- **Real gases deviate from ideal behavior because:**
  - Their molecules have no volume.
  - Their molecules have no intermolecular forces.
  - Their molecular volume and intermolecular forces are significant.
  - Their molecules have no kinetic energy.

- **Which intermolecular force occurs between ions and polar solvent molecules?**
  - London dispersion forces
  - Ion–dipole forces
  - Dipole–dipole forces
  - Covalent bonds

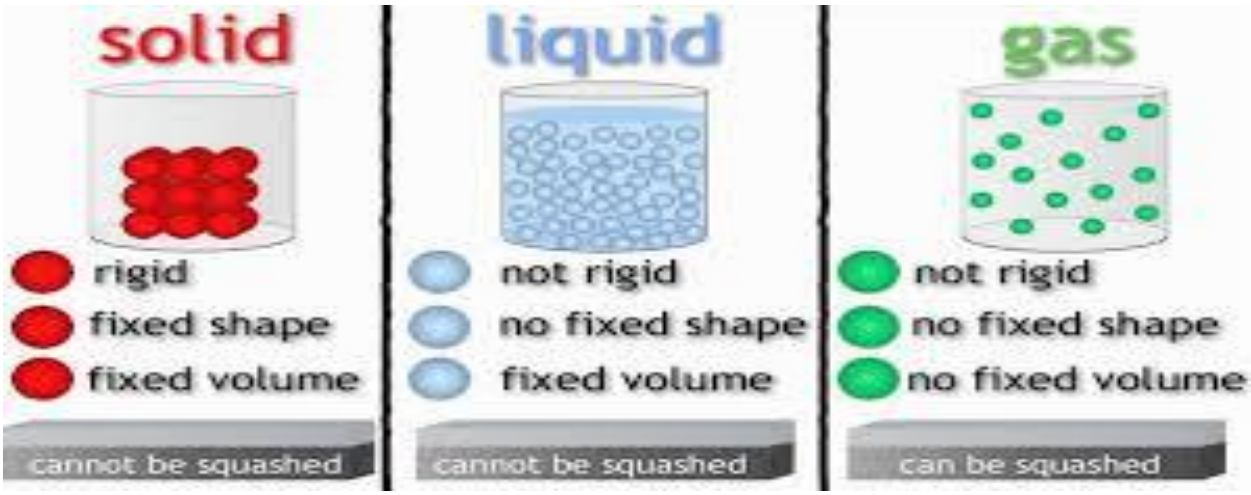
- **Surface tension describes a liquid's resistance to:**
  - Increasing its surface area
  - Increasing its mass
  - Decreasing its density
  - Increasing its volume only

- **The normal boiling point of a pure substance is its boiling temperature at:**
  - 0 atm
  - 1 atm
  - 2 atm
  - Any external pressure

- **Convert Pressure 1150 torr to mmHg.....**
  - 11.50
  - 115.0
  - 1150

- **If the Kelvin temperature of a gas is doubled, the volume of the gas will increase by __**
  - 3
  - 2
  - 0.5
- **……..describe the relationships between a gas's pressure (P), volume (V), temperature (T), and amount (n)**
  - Gas laws
  - Thermodynamics laws
  - Newton' s laws

9. Compare between Solids ,liquids and gases



Chapter 2: Thermodynamics

1. **A gas absorbs 450 Joules of heat from its surroundings. During this process, the gas does 300 Joules of work on the surroundings. What is the change in the internal energy of the gas?**

Answer:

$\Delta U = Q - W$

$= (450) - (300) = 150$ J

2. **A gas is cooled, and 500 J of heat is removed from it. During this process, 800 J of work is done on the gas. What is the change in the internal energy of the gas?**

Answer:

$\Delta U = Q - W$

$= (-500) - (-800) = 300$ J

3. Write (True) or (False) for each of the following statements.

- A path function is a property whose value does not depend on the path to reach that specific function or value (×)
- Diffusion of perfume molecules from one side of a room to the other is a nonspontaneous process **(×)**
- The internal energy of a system depends on the path followed to reach its present state (×)

4. Choose the correct answer:

- **Thermodynamics is the study of**:
  - Chemical bonding only
  - Energy, energy transformations, and their relation to matter
  - Reaction rates only
  - The structure of atoms only
- **Which of the following can exchange both matter and energy with its surroundings?**
  - Open system
  - Closed system
  - Isolated system
  - None of these
- **A spontaneous process proceeds:**
  - Only under continuous external influence
  - On its own without external influence
  - Only at absolute zero
  - Only when heat is absorbed

- **Which thermodynamic law expresses the conservation of energy?**
  - Zeroth law
  - First law
  - Second law
  - Third law

- **If heat is released by a system, the sign of (Q) in the lecture's first-law convention is:**
  - Positive
  - Negative
  - Always zero
  - Equal to the pressure

- **Heat is transferred between two objects because of a difference in:**
  - Volume
  - Temperature
  - Chemical identity
  - Mass only

- **In thermodynamics ..... is not a state variable**
  - Density
  - Entropy
  - Heat

- **The least random state of the water system is:**
  - Ice
  - Liquid water
  - Steam

- **An object has a low temperature, its atoms or molecules move**
  - slowly
  - rapidly
  - not effected

6. Which of the following processes are spontaneous? Which are nonspontaneous?

- Sugar dissolves in water when stirred. (**spontaneous)**
- Heat flows from a hot object to a cold object. (**spontaneous)**
- Perfume molecules diffuse throughout a room.(**spontaneous)**
- Water flows uphill without any external energy input (**nonspontaneous)**
- A gas expands to fill the entire available container.(**spontaneous)**
- Heat flows from a cold object to a hot object without external influence (**nonspontaneous)**

7. **Predict the sign of ΔS**

- $H_2O_{(l)} \rightarrow H_2O_{(g)}$ **Answer:** ΔS >0 — Positive, because liquid water changes into gas, increasing randomness.
- $2KClO_{3(s)} \rightarrow 2KCl(s) + 3O_{2(g)}$ **Answer:** ΔS >0 — Positive, because a gas is produced from solid reactants.
- $2SO_{2(g)} + O_{2(g)} \rightarrow 2SO_{3(g)}$ **Answer:** ΔS < 0— Negative, because the total number of gas moles decreases from 3 to 2.

Chapter 3: Thermochemistry

1. **Calculate $\Delta H°_f$ in kJ/mol for benzene(C_6H_6), from the following data:**

$$\mathbf{2\ C_6H_{6(l)} + 15\ O_{2(g)} \rightarrow 12\ CO_{2(g)} + 6\ H_2O_{(l)} \quad \Delta H°r = -6534\ kJ}$$

$\Delta H°_f (CO_2) = -393.5$ kJ/mol **$\Delta H°_f (H_2O) = -285.8$ kJ/mol**

Answer:

$$\Delta H°r = \sum \Delta H_{◦f\ \text{Products}} - \sum \Delta H°_{f\ \text{Reactant}}$$

$$\Delta H°_r = [\Delta H_{◦f}(CO_2) + \Delta H°_f(H_2O)] - [\Delta H°_f(C_6H_6)]$$

$$-6534\ kJ = [12\ mol * -393.5\ kJ/mol + 6\ mol * -285.8\ kJ/mol] - [2\ mol * \Delta H°_f(C_6H_6)]$$

$$\Delta H°_f(C_6H_6) = 48.6\ kJ/mol$$

+ 1 mol* -635.1 kJ/mol] - [1 mol *-1206.9 kJ/mol]

$$\Delta H°r = 178.3\ kJ/mol$$

2. **Calculate the work done during a reaction where the volume expands from 8.0 L to 11.0 L against an external pressure of 2.0 atm.**

Answer:

$\Delta V = V_f - V_i = 11 - 8 = 3$ L

$W = - P \Delta V$

$W = - 2 \times 3 = - 6$ atm.L

Convert to joules using 1 atm.L = 101.325 J

$W = - 6 \times 101.325 = -608$ J = -0.61 kJ

3. Choose the correct answer:

- **Thermochemistry is concerned primarily with the heat absorbed or released during:**
  - Chemical reactions
  - Changes in position only
  - Atomic structure determination
  - Measurement of pressure only

- **Which of the following is a factor affecting the internal energy of a system?**
  - Chemical identity
  - Temperature
  - Physical state
  - All of the above

- **Chemical energy is classified as a form of:**
  - Potential energy
  - Pressure energy only
  - Heat transfer
  - Volume

- **Which statement correctly describes an endothermic process?**
  - Heat flows out of the system, and ΔH is negative.
  - Heat flows into the system, and ΔH is positive.
  - Heat must always be zero.
  - The system cannot undergo an enthalpy change.

- **Which sign is associated with the enthalpy change of an exothermic process?**
  - Positive
  - Negative
  - Always zero
  - Always equal to one

- **The standard heat of formation represents the enthalpy change associated with forming:**
  - One mole of a substance in its standard state
  - One gram of a substance under any conditions
  - Two moles of any compound
  - One mole of any element from an arbitrary state

4. **Complete the following statements using the appropriate words from the word bank below. Each word should be used only once.**

[Positive. - Expansion (PV) work-.Internal energy- Zero.-Thermodynamics.**]**

- Thermochemistry is a branch of ____**Thermodynamics**_____ that studies heat changes accompanying chemical reactions.
- The sum of the kinetic and potential energies of all molecules or ions in a system is called ___**Internal energy**_____**.**
- The enthalpy change of an endothermic process has a __**Positive**_ sign.
- The standard heat of formation of an element in its standard state is equal to ____**Zero**___.

5. Write (True) or (False) for each of the following statements.

- Internal energy is a state function and does not depend on the system's history (√)
- At higher temperatures, molecules generally move more slowly than at lower temperatures (×)
- Heat is an amount of energy transfer caused by a temperature difference (√)
- An exothermic process absorbs heat from the surroundings and has a positive enthalpy change (×)
- Enthalpy is the sum of internal energy and the product of pressure and volume (√)
  - The internal energy of a system may depend on its chemical identity, sample size, temperature, pressure, and physical state (√)

Chapter 4: Solutions

5. **Molarity: Find the molarity of a solution prepared by dissolving 20 g of NaOH in sufficient water to prepare 500 mL of solution. (Na = 23, O = 16, H = 1)**

   Answer

   Molar mass =23+16+1=40 g/mol

   n = mass / molar mass = 20 / 40 = 0.5 moles

$$\text{Molarity} = \frac{\text{moles solute}}{\text{1 L of solution}}$$

OR

$$M = \frac{\text{mol}}{\text{L}}$$

= 0.5/ 0.5=1

6. **Mass percentage: Determine the mass percentage of NaCl in a solution containing 15 g of NaCl and 135 g of water.**

   Answer

   = [15 / (15+135)] x 100= 10%

7. **Henry's law: The solubility of a gas in a liquid is 0.025 mol/L at a partial pressure of 2 atm. Determine its solubility at 5 atm at the same temperature.**

   Answer

Henry's law $\text{Solubility} = k \cdot P$

$$\frac{s_1}{s_2} = \frac{p_1}{p_2}$$

s_2 = (0.025) x (5/2) = 0.0625 mol/L

1. Choose the correct answer:

- **. A solution is a _____ mixture**.
  - Heterogeneous
  - <u>Homogeneous</u>
  - Mechanical
  - Non-uniform

- **The substance that dissolves another substance is called the:**
  - Solute

- Suspension
- Solvent
- Precipitate

➢ **An aqueous solution is a solution in which the solvent is:**
- Ethanol
- Acetone
- Water
- Benzene

➢ **In a solution containing a gas dissolved in a liquid, the gas is generally considered the:**
- Solvent
- Solute
- Precipitate
- Suspension

➢ **A solution that conducts electricity is classified as a/an:**
- Nonelectrolyte solution
- Electrolyte solution
- Supersaturated solution
- Unsaturated solution

➢ **A solution containing less than the maximum amount of dissolved solute at a given temperature and pressure is**:
- Saturated
- Supersaturated
- Unsaturated
- Concentrated

➢ **A solution containing more dissolved solute than the normal maximum amount at a given temperature is called:**
- Saturated
- Unsaturated
- Dilute
- Supersaturated

➢ **Which factor significantly affects the solubility of a solid in a liquid?**
- Temperature
- Container shape
- Colour
- Surface appearance

➢ **As the temperature increases, the solubility of gases in liquids:**
- Increases
- Decreases
- Remains constant
- Becomes unlimited

2. Complete the following statements using the appropriate scientific term from the word bank below. Each word should be used only once.

[– Solute – Alloys - Solvent – Aqueous – Nonelectrolyte – Saturated – Decreases – Partial - Homogeneous– Negligible]

- A solution is a ____**Homogeneous**_____ mixture of two or more substances.
- The substance that is dissolved in a solution is called the __**Solute**___
- The component that usually determines the physical state of a solution is the ___**Solvent**_____**.**
- A solution in which water acts as the solvent is called an ____**Aqueous**_____ solution.
- Solid solutions of metals are known as _____**Alloys** _____**.**
- Solutions that do not conduct electric current are called ____**Nonelectrolyte**_____ solutions.
- A solution containing the maximum amount of solute that can dissolve under specified conditions is called a ___**Saturated**_____ solution.
- The solubility of gases in liquids generally _____**Decreases**_____ as temperature increases.
- The solubility of a gas in a liquid increases as the gas's _____**Partial**_____ pressure increases at constant temperature.
- The effect of pressure on the solubility of solids is practically ___Negligible

3. Write True or False for each statement.

- Alloys are examples of solid solutions (√)
- Electrolyte solutions conduct electricity (√)
- An unsaturated solution contains more solute than the maximum amount that can dissolve at a given temperature (×)
- The solubility of a solid in a liquid affected by temperature (√)
- Increasing temperature generally decreases the solubility of gases in liquids (√)
- Pressure has practically no effect on the solubility of gases in liquids (×)
- Henry's law relates gas solubility to the partial pressure of the gas above the solution (√)

4. Answer the following questions

- What is an aqueous solution?

  An aqueous solution is a solution in which water acts as the solvent.

- Classify solutions according to their electrical conductivity.

  Electrolyte solutions and non-electrolyte solutions

- Explain why alloys are considered solid solutions.

  Because they consist of two or more elements mixed uniformly in the solid state.

- Describe the effect of temperature on the solubility of solids in liquids.

  The solubility of solids increases as temperature increases.

- Explain the effect of temperature on the solubility of gases in liquids.

  Temperature is inversely proportional to the solubility of gases in liquids.

- State Henry's law.



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